

EDUCATION

The University of Chicago

Bachelor of Sciences: Geophysical Sciences, Minor: Astrophysics

Cumulative GPA: 3.4/4.0

Course Highlights: Intro. To Python Prog. w/ Appl. to Astro Statistics, Phylogenetics in the Fossil Record, Comparative Vert. Anatomy, Dinosaur Science, Intro. to Invert. Paleobiology, Physical Geology, Earth History, Biodiversity, Biogeochemical Cycles, Geobiology

Awards, Honors, Other: QuestBridge NCM Finalist, Odyssey Scholar, National Merit Scholar 2021-2025

Chicago, IL

Jun. 2025

RELEVANT EXPERIENCE

Slater Lab – Multivariate Ecology on Mammalian Herbivores

Research Assistant

Chicago, IL

Jun. 2024 – Current

- Reviewed literature on available dietary studies on mammalian herbivores, particularly artiodactyls, for use in a Bayesian prediction of multivariate ecology from dental phenotypic data – with the goal of eventual use on extinct taxa – and collaborated with peers on adapting Carnivoran-based classifications in Wisniewski (2023) for herbivorous clades
- Gaining experience 3D scanning of Field Museum specimens and processing in R

Eodiscid Trilobite Fossil Processing & Morphological Analysis

Metcalf Intern

Chicago, IL

Feb. 2024 – Current

- Cleaned, stained, and photographed dozens of mm-scale eodiscid trilobites for morphological analysis and use in biostratigraphy

Sereno Fossil Lab

Volunteer, Summer Student Lab Technician

Chicago, IL

Sept. 2021 – Dec. 2024, Summer Tech Jun. – Aug. 2023

- Prepared fossil vertebrate specimens, as well as creating visualizations, diagrams, and casts for display, outreach, and donor gifts
- Participated in small, week-long expedition to exposed Morrison, Cloverly formations at base of Bighorn Mountains, involving excavating a *Camarasaurus* vertebra, using B72 for field stabilization, etc.
- Presented the role of art in paleontology to members of the local community during lab open houses

Liew House Dorm Resident Assistant

Senior RA as of Sept. 2024

Chicago, IL

Sept. 2023 – Jun. 2025

- Fostered a positive living environment for residents, resolving conflicts, and organizing biweekly social events.

Deep Skies Lab

Metcalf Intern

Chicago, IL

Feb. 2023 – Oct. 2023

- Utilized simulation-based-inference to analyze the relationships between cosmological constraints, dark matter halos in galaxy clusters, and the observed SZ effect in the cosmic microwave background
- Used regression and classification machine learning techniques in Python in applications to cosmology and astrophysics, particularly gaining proficiency in GitHub and numpy, pandas, scipy, astropy, matplotlib, keras packages
- Participated in presentation of research to members of Deep Skies and Fermilab

ACTIVITIES & RESEARCH

Pterosaur Phylogeny

Dec. 2022 – Dec. 2023

- Performed an informal literature review of pterosaur crest morphology and phylogeny, particularly that of problem clades (anurognathidae), and quantitatively analyzed public morphological datasets. Parsimony- and Bayesian-based trees were produced from these same datasets and compared with the goal of finding a consensus result amidst consistently competing hypotheses
- Gained experience in comparative cladistics and with popular paleontology packages like ape, strap, phytools, as well as programs like RevBayes, PAUP, TNT, and Mesquite. Developed pipeline to process PBDB data

Severn Fellows Program

Severn Fellow

Severna Park, MD

May, 2020 – May, 2021

- Drafted, proposed, budgeted, and planned independent research and explorational project
- Researched multiple theropod dinosaur species to understand phylogeny, ecology, appearance, and anatomy to compile data for an informative and visually interesting series of pages and paleoart.

UChicago Paleontology Club

Board Member, Co-President

Chicago, IL

Dec. 2022 – Jun. 2025

- Club fosters a community of paleontologists, enthusiasts by developing an online presence, hosting talks, and leading workshops

SKILLS & INTERESTS

Data, Programming & Computing – Python, R, RevBayes, Mesquite, PAUP, TNT, Microsoft Office (Word, Excel, PowerPoint), Adobe (Photoshop, Illustrator) LaTeX. Specimen Light Scanning & Processing.

Other – Art (Pencil, Pastel, Digital), Competitive Ballroom and Latin Dance, Rockclimbing, Gardening (w/ Phoenix Farms)